

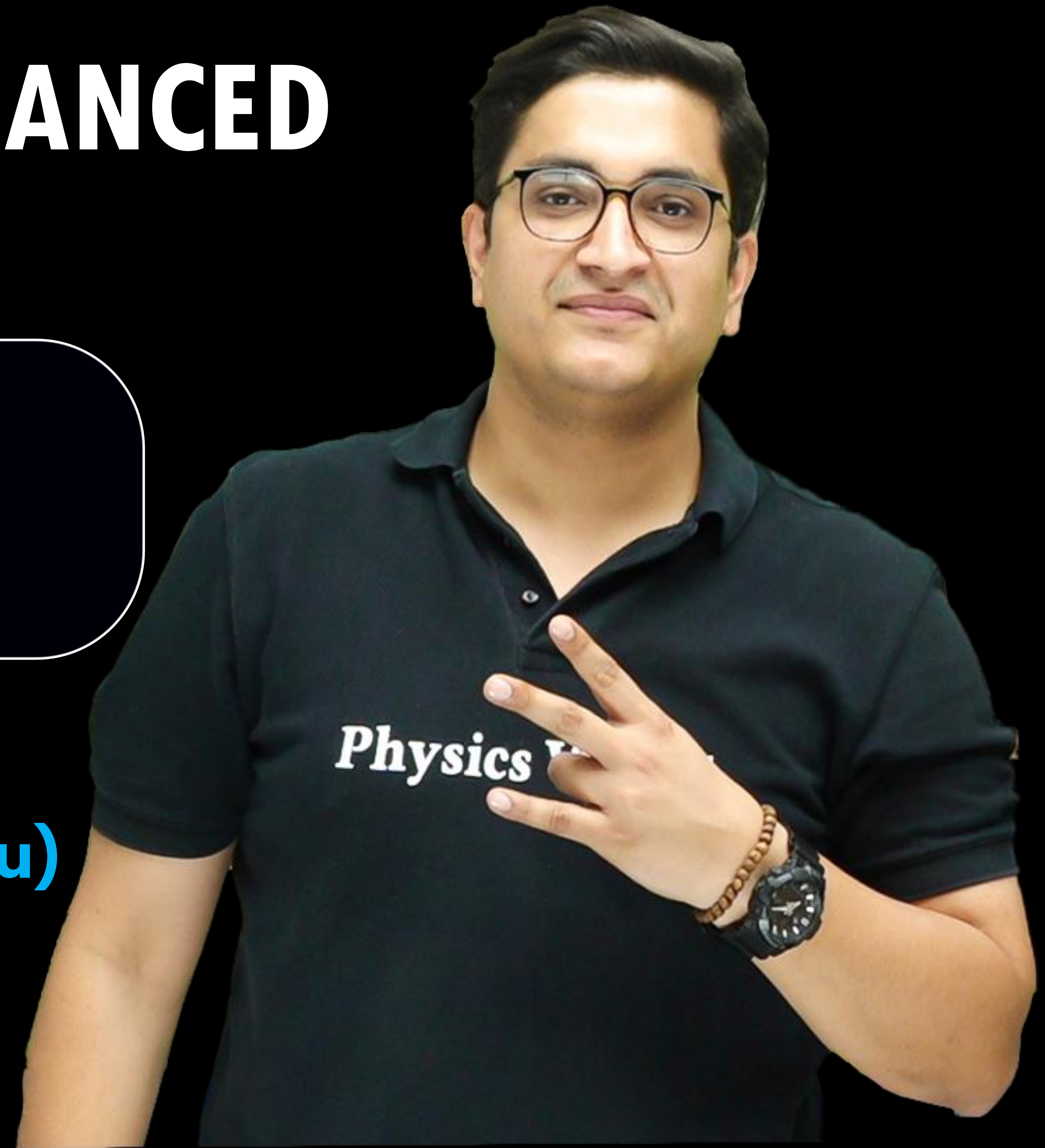
JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

STATISTICS

DIBY (DO IT BY YOURSELF)

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DIBY-01

The mean age of 25 teachers in a school is 40 years. A teacher retires at the age of 60 years and a new teacher is appointed in his place. If the mean age of the teachers in this school now is 39 years, then the age (in years) of the newly appointed teacher is:

[JEE Mains-2021]

[Ans. : 35]

DIBY-02

Let the mean and the standard deviation of the observation 2, 3, 3, 4, 5, 7, a , b be 4 and $\sqrt{2}$ respectively. Then the mean deviation about the mode of these observations is :

[JEE Mains-2025]

- (A) 1 (b) $\frac{3}{4}$ (c) 2 (d) $\frac{1}{2}$

[Ans. : (A)]

DIBY-03

Let $x_1, x_2, x_3, \dots, x_{20}$ be in geometric progression with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data is constructed replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal to $\bar{x}$ is

[JEE Mains-2022]

[Ans. : 142]

DIBY-04

Let μ be the mean and σ be the standard deviation of the distribution

x_1	0	1	2	3	4	5
f_1	$k + 2$	$2k$	$k^2 - 1$	$k^2 - 1$	$k^2 + 1$	$k - 3$

where $\sum f_i = 62$. if $[x]$ denotes the greatest integer $\leq x$, then $[\mu^2 + \sigma^2]$ is equal

[JEE Mains-2023]

- (A) 8 (B) 7 (C) 6 (D) 9

[Ans. : (A)]

DIBY-05

The mean of 10 numbers $7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14, \dots$ is _____.

[JEE Mains-2021]

[Ans. : 398]

DIBY-06

Let the median and the mean deviation about the median of 7 observation

$170, 125, 230, 190, 210, a, b$ be 170 and $\frac{205}{7}$ respectively. Then the mean deviation about the mean of these 7 observations is:

[JEE Mains-2024]

- (A) 31 (B) 28 (C) 30 (D) 32

[Ans. : (C)]

DIBY-07

Let $a, b \in R$. Let the mean and the variance of 6 observations $-3, 4, 7, -6, a, b$ be 2 and 23, respectively. The mean deviation about the mean of these 6 observations is:

[JEE Mains-2024]

- (A) $\frac{13}{3}$ (B) $\frac{16}{3}$ (C) $\frac{11}{3}$ (D) $\frac{14}{3}$

[Ans. : (A)]

DIBY-08

Let $a, b, c \in N$ and $a < b < c$. Let the mean, the mean deviation about the mean and the variance of the 5 observations $9, 25, a, b, c$ be 18, 4 and $\frac{136}{5}$, respectively. Then $2a + b - c$ is equal to ____.

[JEE Mains-2024]

[Ans. : 33]

DIBY-09

Let S be the set of all values of a_1 for which the mean deviation about the mean of 100 consecutive positive integers $a_1, a_2, a_3, \dots, a_{100}$ is 25. Then S is

[JEE Mains-2023]

- (A) ϕ (B) $\{99\}$ (C) N (D) $\{9\}$

[Ans. : (C)]

DIBY-10

The frequency distribution of the age of students in a class of 40 students is given below.

Age	15	16	17	18	19	20
No. of Students	5	8	5	12	x	y

If the mean deviation about the median is 1.25, then $4x + 5y$ is equal to :

- (A) 43 (B) 44 (C) 47 (D) 46

[JEE Mains-2024]

[Ans. : (B)]

DIBY-11

Let the mean of the data

x	1	3	5	7	9
Frequency (f)	4	24	28	α	8

be 5. If m and σ^2 are respectively the mean deviation about the mean and the variance of the data, then $\frac{3\alpha}{m+\sigma^2}$ is equal to ____

[JEE Mains-2023]

[Ans. : 8]

DIBY-12

If the mean and the variance of $6, 4, a, 8, b, 12, 10, 13$ are 9 and 9.25 respectively, then $a + b + ab$ is equal to:

[JEE Mains-2025]

- (A) 105 (B) 103 (C) 100 (D) 106

[Ans.: (B)]

DIBY-13

Let the Mean and Variance of five observations $x_1 = 1, x_2 = 3, x_3 = a, x_4 = 7$ and $x_5 = b, a > b$, be 5 and 10 respectively. Then the Variance of the observations $n + x_n, n = 1, 2, 5$ is

[JEE Mains-2025]

- (A) 17 (B) 16.4 (C) 17.4 (D) 16

[Ans.: (D)]

DIBY-14

The mean and standard deviation of 100 observations are 40 and 5.1, respectively. By mistake one observation is taken as 50 instead of 40. If the correct mean and the correct standard deviation are μ and σ respectively, then $10(\mu + \sigma)$ is equal to

[JEE Mains-2025]

- (A) 445 (B) 451 (C) 447 (D) 449

[Ans.: (D)]

DIBY-15

The variance of the numbers $8, 21, 34, 47, \dots, 320$ is ____.

[JEE Mains-2025]

[Ans. : 8788]

DIBY-16

For a statistical data $x_1, x_2, \dots, x_{10}$ of 10 values, a student obtained the mean as 5.5 and $\sum_{i=1}^{10} x_i^2 = 371$. He later found that he had noted two values in the data incorrectly as 4 and 5, instead of the correct values 6 and 8, respectively. The variance of the corrected data is

[JEE Mains-2025]

- (A) 7 (B) 5 (C) 9 (D) 4

[Ans. : (A)]

DIBY-17

Let the mean and standard deviation of marks of class A of 100 students be respectively 40 and $\alpha(> 0)$, and the mean and standard deviation of marks of class B of n students be respectively 55 and $30 - \alpha$. If the mean and variance of the marks of the combined class of $100 + n$ students are respectively 50 and 350, then the sum of variances of classes A and B is:

[JEE Mains-2023]

- (A) 500 (B) 650 (C) 450 (D) 900

[Ans. : (A)]

DIBY-18

Let the mean and variance of 8 numbers $x, y, 10, 12, 6, 12, 4, 8$ be 9 and 9.25 respectively. If $x > y$, then $3x - 2y$ is equal to

[JEE Mains-2023]

[Ans. : 25]

DIBY-19

Let the positive numbers a_1, a_2, a_3, a_4 and a_5 be in a G.P. Let their mean and variance be $31/10$ and m/n respectively, where m and n are co-prime. If the mean of their reciprocals is $31/40$ and $a_3 + a_4 + a_5 = 14$, then $m + n$ is equal to

[JEE Mains-2023]

[Ans. : 211]

DIBY-20

If the variance of the frequency distribution

x_i	2	3	4	5	6	7	8
Frequency f_i	3	6	16	α	9	5	6

is 3, then α is equal to

[JEE Mains-2023]

[Ans. : 5]

DIBY-21

Let the six numbers $a_1, a_2, a_3, a_4, a_5, a_6$ be in A.P. and $a_1 + a_3 = 10$. If the mean of these six numbers is $\frac{19}{2}$ and their variance is σ^2 , then $8\sigma^2$ is equal to **[JEE Mains-2023]**

- (A) 220 (B) 210 (C) 200 (D) 105

[Ans.: (B)]

DIBY-22

The number of value of $a \in N$ such that the variance of $3, 7, 12, a, 43 - a$ is a natural number is: **[JEE Mains-2022]**

- (A) 0 (B) 2 (C) 5 (D) Infinite

[Ans.: (B)]

DIBY-23

Let the mean and variance of the frequency distribution

x	$x_1 = 2$	$x_2 = 6$	$x_3 = 8$	$x_4 = 9$
f	4	4	α	β

be 6 and 6.8 respectively. If x_3 is changed from 8 to 7, then the mean for the new data will be: **[JEE Mains-2021]**

- (A) 4 (B) 5 (C) $\frac{17}{3}$ (D) $\frac{16}{3}$

[Ans. (c)]



DIBY-24

Let $x_1, x_2, \dots, x_{18}$ be eighteen observations such that $\sum_{i=1}^{18} (x_i - \alpha) = 36$ and $\sum_{i=1}^{18} (x_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha - \beta|$ is.....

[JEE Mains-2021]

[Ans. : 4]

DIBY-25

An online exam is attempted by 50 candidates out of which 20 are boys. The average marks obtained by boys is 12 with a variance 2. The variance of marks obtained by 30 girls is also 2. The average marks of all 50 candidates is 15. If μ is the average marks of girls and σ^2 is the variance of marks of 50 candidates, then $\mu + \sigma^2$ is equal to

[JEE Mains-2021]

[Ans. : 23]

DIBY-26

If the variance of the terms in an increasing A.P., $b_1, b_2, b_3, \dots, b_{11}$ is 90, then the common difference of this A.P. is ____

[JEE Mains-2020]

[Ans. : 3]



DIBY-27

If the variance of the first n natural numbers is 10 and the variance of the first m even natural numbers is 16, then $(m + n)$ is equal to ____

[JEE Mains-2020]

[Ans. : 18]

DIBY-28

A student scores the following marks in five tests: 45, 54, 41, 57, 43. His score is not known for the sixth test. If the mean score is 48 in the six tests, then the standard deviation of the marks in six tests is

[JEE Mains-2019]

(a) $\frac{10}{\sqrt{3}}$

(B) $\frac{100}{\sqrt{3}}$

(C) $\frac{100}{3}$

(D) $\frac{10}{3}$

[Ans. (A)]

DIBY-29

A data consists of n observations: $x_1, x_2, \dots, x_n$. If $\sum_{i=1}^n (x_i + 1)^2 = 9n$ and $\sum_{i=1}^n (x_i - 1)^2 = 5n$, then the standard deviation of this data is:

[JEE Mains-2019]

(A) 5

(B) $\sqrt{5}$

(C) $\sqrt{7}$

(D) 2

[Ans. : (B)]

DIBY-30

If both the mean and the standard deviation of 50 observations $x_1, x_2, \dots, x_{50}$ are equal to 16, then the mean of $(x_1 - 4)^2, (x_2 - 4)^2, \dots, (x_{50} - 4)^2$ is: **[JEE Mains-2019]**

- (A) 525 (B) 380 (C) 480 (D) 400

[Ans. : (D)]

DIBY-31

Consider a set of $3n$ numbers having variance 4. In this set, the mean of first $2n$ numbers is 6 and the mean of the remaining n number is 3. A new set is constructed by adding 1 into each of first $2n$ numbers, and subtracting 1 from each of the remaining n numbers. If the variance of the new set is k , then $9k$ is equal to ____ . **[JEE Mains-2021]**

[Ans. : 5]

DIBY-32

The mean and variance of a set of 15 numbers are 12 and 14 respectively. The mean and variance of another set of 15 numbers are 14 and σ^2 respectively. If the variance of all the 30 numbers in the two sets is 13, then σ^2 is equal to **[JEE Mains-2023]**

- (A) 9 (B) 12 (C) 11 (D) 10

[Ans. : (D)]

DIBY-33

Let $X = \{11, 12, 13, \dots, 40, 41\}$ and $Y = \{61, 62, 63, \dots, 90, 91\}$ be the two sets of observations. If $\bar{x}$ and $\bar{y}$ are their respective means and σ^2 is the variance of all the observations in $X \cup Y$, then $|\bar{x} + \bar{y} - \sigma^2|$ is equal to ____.

[JEE Mains-2023]

[Ans. : 603]

DIBY-34

If the mean and variance of eight numbers $3, 7, 9, 12, 13, 20, x$ and y be 10 and 25 respectively, then $x \cdot y$ is equal to ____.

[JEE Mains-2020]

[Ans. : 54]



Consider the following frequency distribution:

Value	4	5	8	9	6	12	11
Frequency	5	f_1	f_2	2	1	1	3

Suppose that the sum of the frequencies is 19 and the median of this frequency distribution is 6.

For the given frequency distribution, let α denote the mean deviation about the mean, β denote the mean deviation about the median, and σ^2 denote the variance.

Match each entry in List-I to the correct entry in List-II and choose the correct option.

- A

I → T; II → R; III → Q; IV → S
- B

I → T; II → Q; III → R; IV → P
- C

I → T; II → R; III → Q; IV → P
- D

I → R; II → Q; III → T; IV → S

List-I		List-II	
I.	$7f_1 + 9f_2$ is equal to	P.	146
II.	19α is equal to	Q.	47
III.	19β is equal to	R.	48
IV.	$19\sigma^2$ is equal to	S.	145
		T.	55



*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**